

Week 5 - Cloud and API deployment

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Batch code: LISUM49

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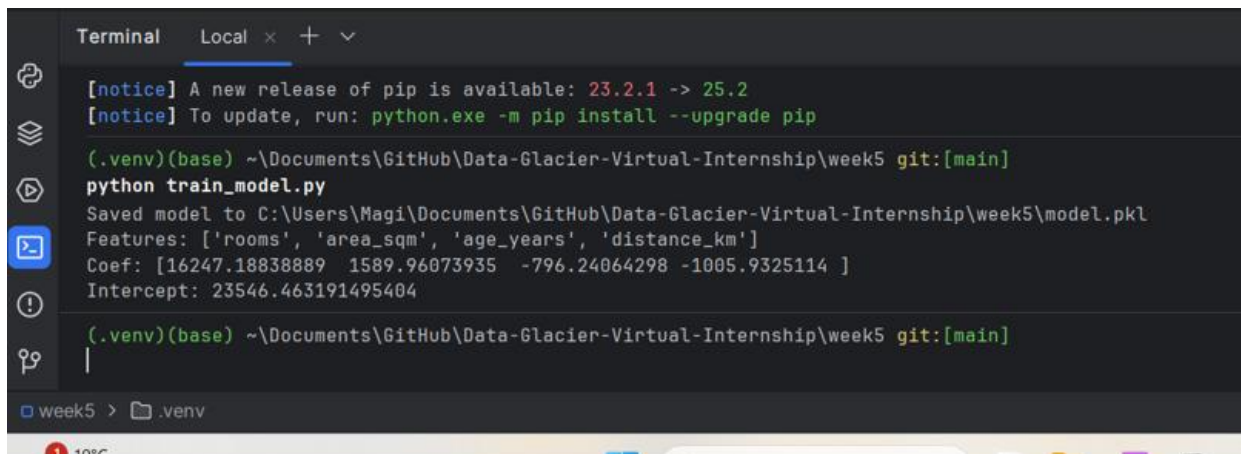
Submitted to: N/A

Introduction

The objective of this project was to train a simple House Price Prediction model using toy data (Week 4 dataset) and deploy it on the cloud. The deployment was implemented using AWS (EC2, ECR, IAM) and automated through GitHub Actions CI/CD.

Step 1: Model Training

The model was trained on toy data and saved as model.pkl.

A terminal window titled 'Terminal' with tabs 'Local' and '+'. It shows the execution of a Python script to train a model. The output indicates the model is saved to a file and lists the features and coefficients.

```
Terminal Local x + v
[notice] A new release of pip is available: 23.2.1 -> 25.2
[notice] To update, run: python.exe -m pip install --upgrade pip

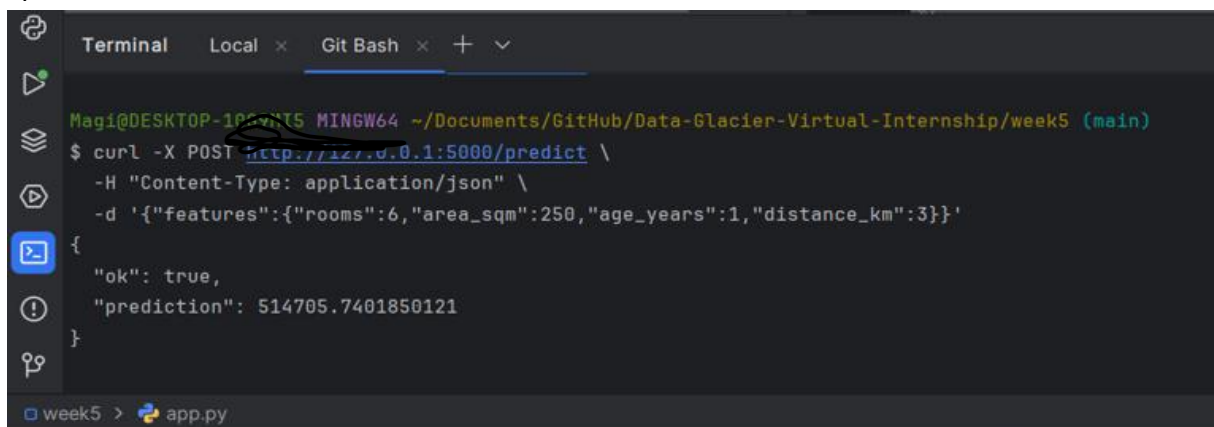
(.venv)(base) ~\Documents\GitHub\Data-Glacier-Virtual-Internship\week5 git:[main]
python train_model.py
Saved model to C:\Users\Magi\Documents\GitHub\Data-Glacier-Virtual-Internship\week5\model.pkl
Features: ['rooms', 'area_sqm', 'age_years', 'distance_km']
Coef: [16247.18838889 1589.96073935 -796.24064298 -1005.9325114 ]
Intercept: 23546.463191495404

(.venv)(base) ~\Documents\GitHub\Data-Glacier-Virtual-Internship\week5 git:[main]
|

week5 > .venv
```

Step 2: Local API Test (JSON)

The API endpoint was tested locally using a curl command to verify predictions via JSON requests.

A terminal window titled 'Terminal' with tabs 'Local' and 'Git Bash'. It shows a curl command being used to test a local API endpoint. The output is a JSON response indicating a successful prediction.

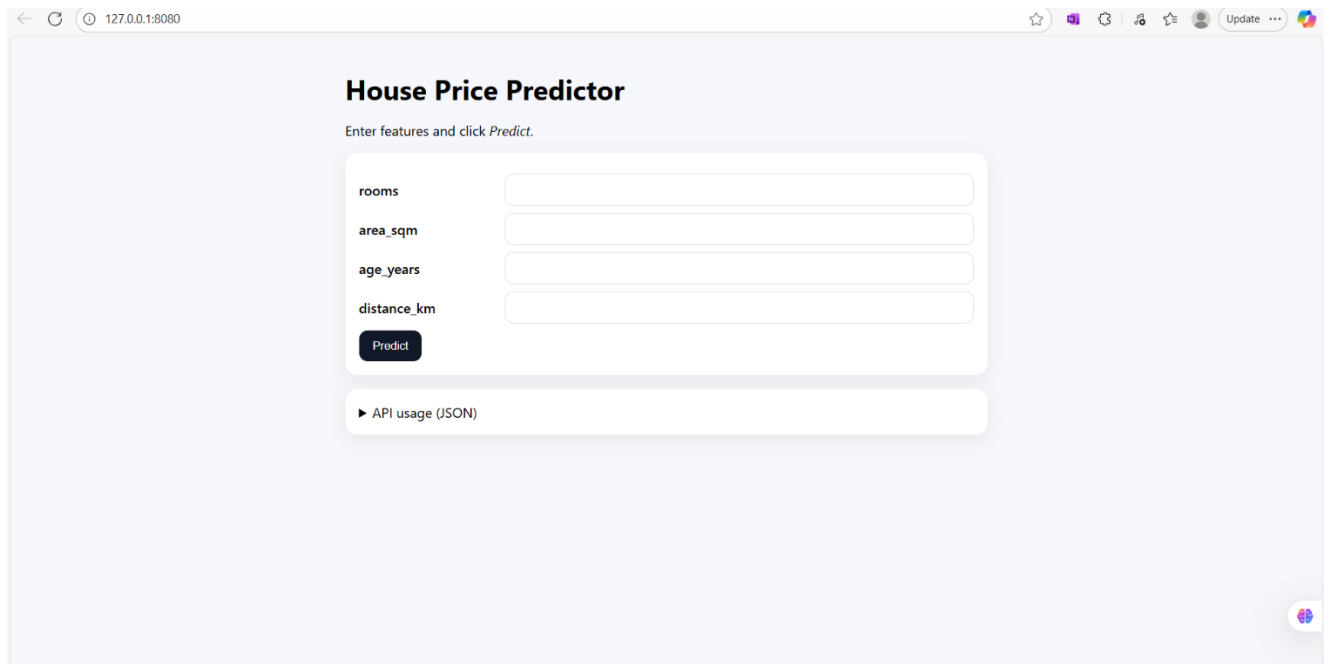
```
Terminal Local x Git Bash x + v
Magi@DESKTOP-102VNT5 MINGW64 ~/Documents/GitHub/Data-Glacier-Virtual-Internship/week5 (main)
$ curl -X POST http://127.0.0.1:5000/predict \
-H "Content-Type: application/json" \
-d '{"features":{"rooms":6,"area_sqm":250,"age_years":1,"distance_km":3}}'

{
  "ok": true,
  "prediction": 514705.7401850121
}

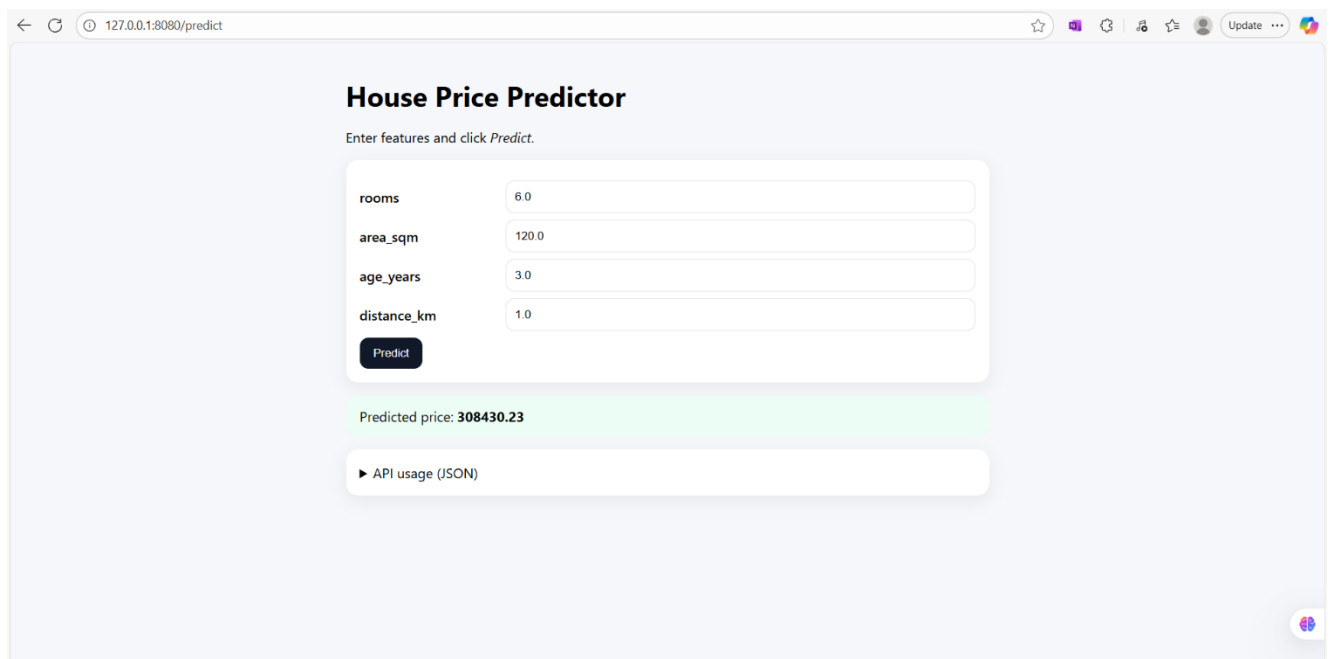
week5 > app.py
```

Step 3: Local Web Application

The Flask-based web application was run locally to test predictions from the user interface.



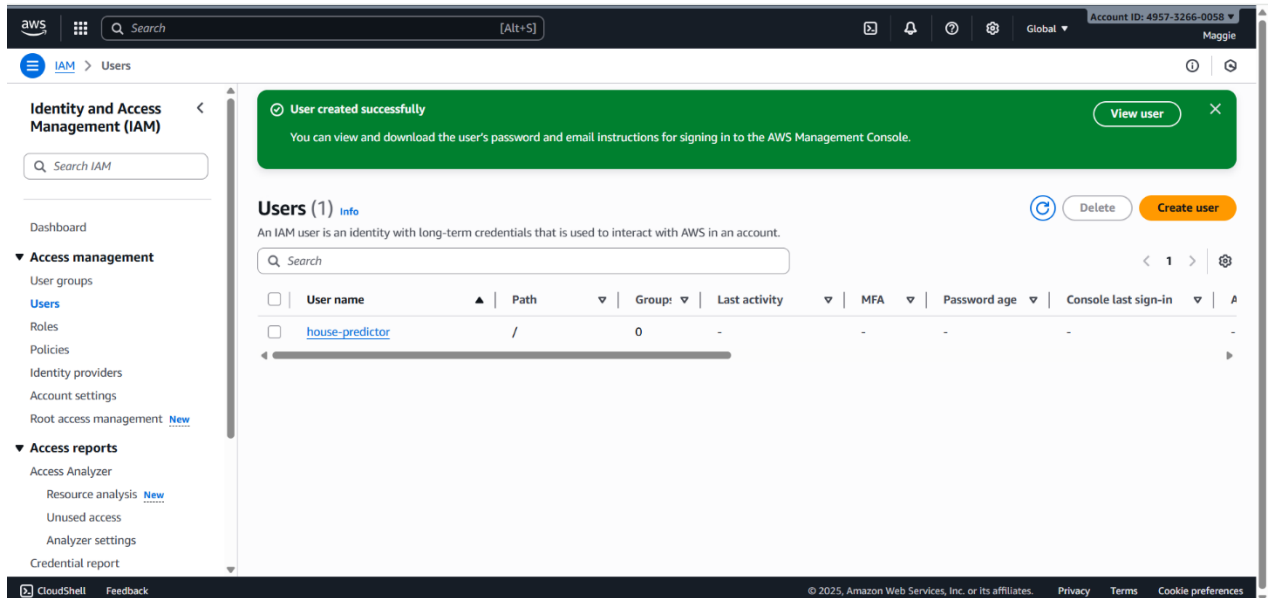
A screenshot of a web browser displaying the 'House Price Predictor' application. The browser's address bar shows '127.0.0.1:8080'. The page has a light blue background. At the top, the title 'House Price Predictor' is displayed in bold black text, followed by the instruction 'Enter features and click Predict.' Below this, there is a white form with four input fields labeled 'rooms', 'area_sqm', 'age_years', and 'distance_km'. A dark blue 'Predict' button is located below the input fields. At the bottom of the form, there is a link that says '► API usage (JSON)'. The browser's toolbar at the top includes back, forward, and refresh buttons, as well as an 'Update' button.



A screenshot of the same 'House Price Predictor' web application, but now showing a prediction result. The browser's address bar shows '127.0.0.1:8080/predict'. The form fields are filled with the values: 'rooms' is 6.0, 'area_sqm' is 120.0, 'age_years' is 3.0, and 'distance_km' is 1.0. The 'Predict' button is still present. Below the form, a green banner displays the 'Predicted price: 308430.23'. At the bottom of the form, there is a link that says '► API usage (JSON)'. The browser's toolbar at the top includes back, forward, and refresh buttons, as well as an 'Update' button.

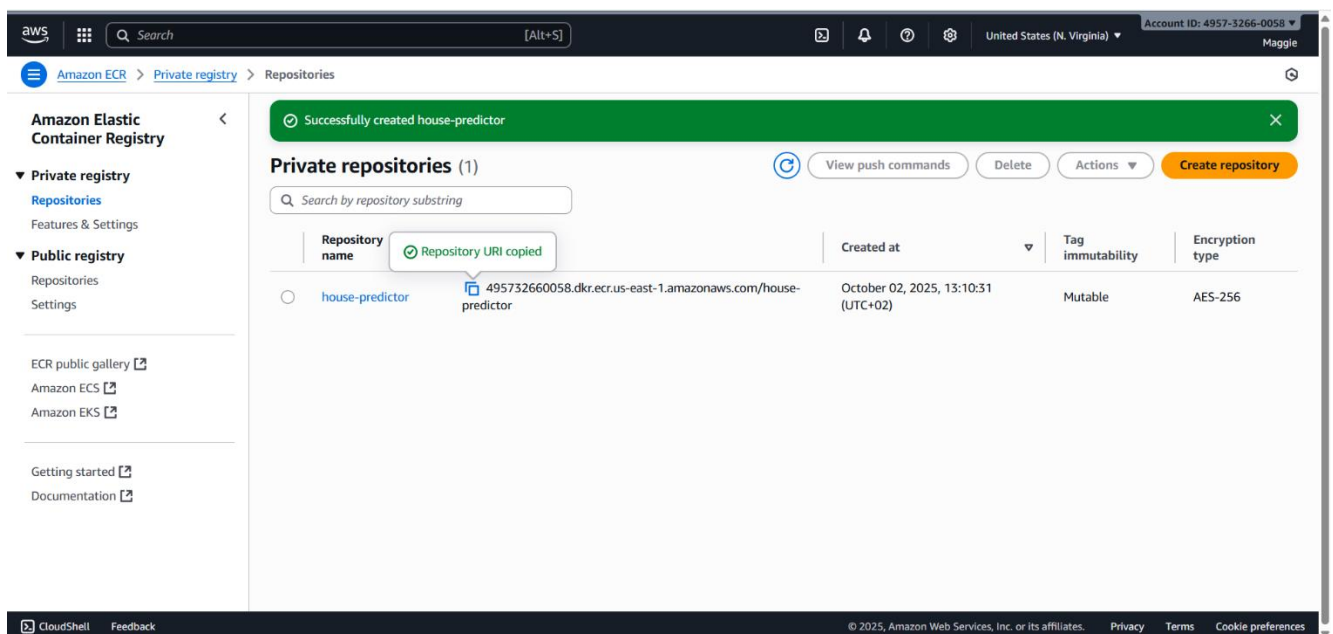
Step 4: AWS IAM Configuration

An IAM user was created with permissions to push Docker images to ECR and enable deployment.



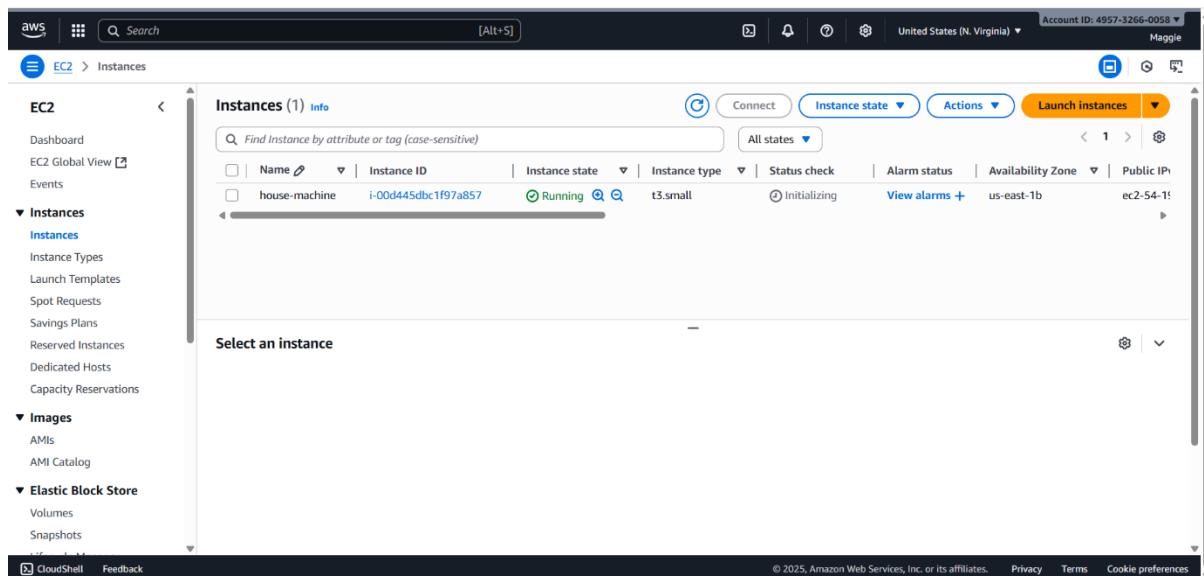
Step 5: AWS Elastic Container Registry

The Docker image for the application was pushed and stored in AWS ECR.



Step 6: AWS EC2 Instance

An EC2 instance was created to host the deployed Docker container.



Step 7: Continuous Integration / Deployment (GitHub Actions)

A CI/CD pipeline was configured with GitHub Actions to automate testing and deployment. (.github/workflows)



magdalena-ivanova-ds

Update app.py ✓

b15a7bc · 1 hour ago

📦 4 Commits

📁 .github/workflows	Update cicd.yaml	1 hour ago
📁 data	Upload app	1 hour ago
📁 static	Upload app	1 hour ago
📁 templates	Upload app	1 hour ago
📁 tests	Upload app	1 hour ago
📄 .gitignore	Upload app	1 hour ago
📄 Dockerfile	Upload app	1 hour ago
📄 README_improved.md	Upload app	1 hour ago
📄 app.py	Update app.py	1 hour ago
📄 cloudrun_deploy.sh	Upload app	1 hour ago
📄 deployment_report_template.md	Upload app	1 hour ago
📄 model.pkl	Upload app	1 hour ago
📄 openapi.yaml	Upload app	1 hour ago
📄 render.yaml	Upload app	1 hour ago
📄 report_deployment_steps.pdf	Upload app	1 hour ago
📄 requirements.txt	Upload app	1 hour ago

No description, website, or topics provided.

🏠 Activity

☆ 0 stars

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Languages

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HTML 20.1%

Dockerfile 8.9%

CSS 24.6%

Shell 11.0%

magdalena-ivanova-ds / House-Price-Predictor

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← Deploy Application Docker Image to EC2 instance

Update ccd.yaml #3 Re-run all jobs

Summary

Jobs

- Continuous-Integration
- Continuous-Deployment

Run details

- Usage
- Workflow file

Triggered via push 3 minutes ago

magdalena-ivanova-ds pushed 8425aa1 main

Status Success

Total duration 1m 9s

Artifacts -

ccd.yaml

on: push

Continuous-Integration 43s

Continuous-Deployment 23s

Step 8: Deployment on AWS EC2

The model and web application were deployed on AWS EC2 using Docker.

54.196.215.13:8080

House Price Predictor

Enter features and click Predict.

rooms

area_sqm

age_years

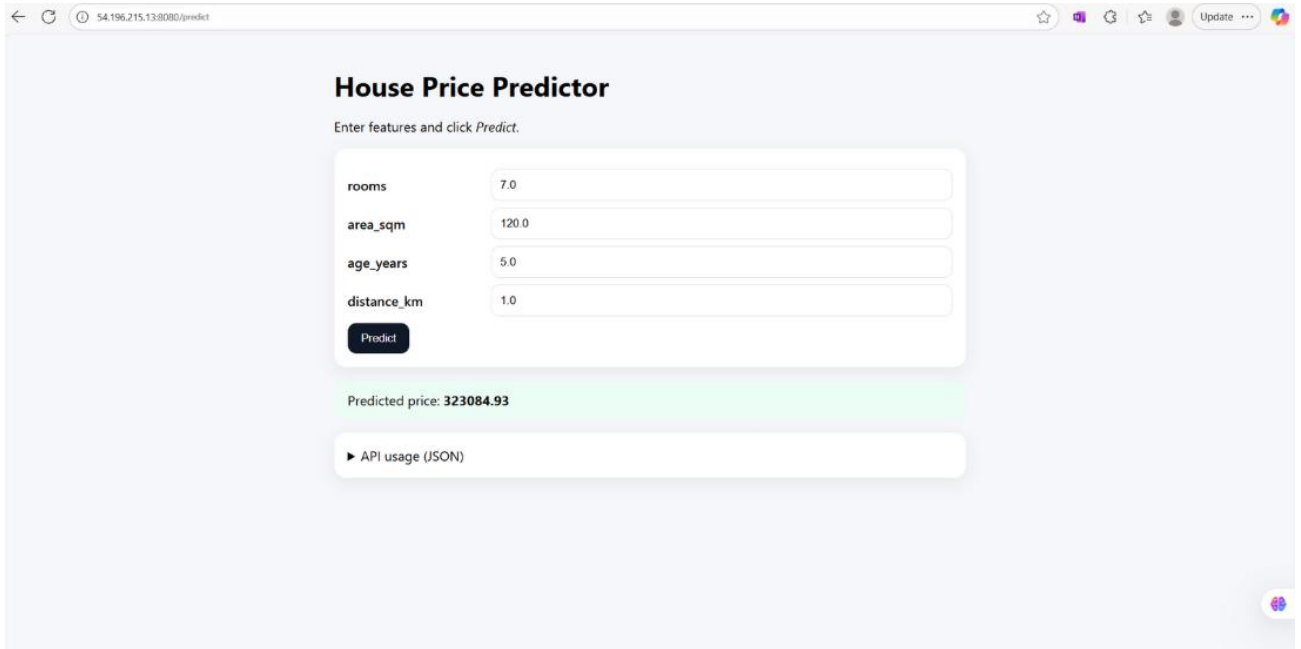
distance_km

Predict

API usage (JSON)

Step 9: Deployed Web Prediction

The deployed application was accessed on the web, and predictions were successfully generated.



The screenshot shows a web browser window with the address bar displaying "54.196.215.138:8080/predict". The page title is "House Price Predictor". Below the title, it says "Enter features and click Predict." There is a form with four input fields: "rooms" (7.0), "area_sqm" (120.0), "age_years" (5.0), and "distance_km" (1.0). A "Predict" button is located below the form. Below the button, the predicted price is displayed: "Predicted price: 323084.93". At the bottom, there is a link to "API usage (JSON)".

Tools Used

- **Python 3.12** – Model training and API development
- **Flask** – Web framework for API and web app
- **Docker** – Containerization of the application
- **AWS EC2** – Cloud compute instance for deployment
- **AWS ECR** – Container registry for storing Docker images
- **AWS IAM** – Secure access management
- **GitHub Actions** – CI/CD automation

Conclusion

The model was successfully trained, tested locally, containerized, and deployed on AWS EC2 using Docker. GitHub Actions was integrated to enable continuous integration and deployment.